Shannon Entropy

math-foundations / concept 36

1 Motivation

Self-information $I(x) = -\log p(x)$ measures the surprise of a single outcome. Shannon entropy is the *expected* self-information: the average surprise we feel when drawing from p repeatedly. Equivalently, it is the optimal expected length (in bits, nats, hartleys, depending on the log base) of a prefix-free code for the random variable.

2 Definitions

Definition 1 (Shannon entropy, discrete)**.** Let X be a discrete random variable on a countable alphabet $\mathcal{X}$ with probability mass function p. The *Shannon entropy* of X is

$$H(X) \;=\; \mathbb{E}[-\log p(X)] \;=\; -\sum_{x\in\mathcal{X}} p(x)\log p(x),$$

with the convention $0\log 0 = 0$ (justified by $\lim_{p\to 0^+} p\log p = 0$).

Definition 2 (Differential entropy, continuous)**.** Let X be a continuous random variable with density f. The *differential entropy* of X is

$$h(X) \;=\; -\int_{\mathbb{R}} f(x)\log f(x)\,dx,$$

whenever the integral exists. Unlike discrete entropy, $h(X)$ may be negative and is not invariant under change of variables.

The base of the logarithm fixes the units: $\log_2 \Rightarrow$ bits; $\ln \Rightarrow$ nats; $\log_{10} \Rightarrow$ hartleys. We use log to denote a generic base unless specified.

3 Basic properties

Lemma 3 (Non-negativity)**.** *For any discrete random variable X, $H(X) \geq 0$, with equality if and only if X is deterministic (i.e., some outcome has probability 1).*

Proof. Each summand $-p(x)\log p(x)$ is non-negative because $0 \leq p(x) \leq 1$ implies $\log p(x) \leq 0$. Equality in the sum requires every term to vanish, which forces $p(x) \in \{0, 1\}$ for all x. $\square$

4 Maximum-entropy theorem

The following result is the central inequality of this lesson.

Theorem 4 (Uniform maximises entropy)**.** *Let X be a discrete random variable on a finite alphabet $\mathcal{X}$ with $|\mathcal{X}| = n$. Then*

$$H(X) \;\leq\; \log n,$$

with equality if and only if X is uniform on $\mathcal{X}$ (i.e., $p(x) = 1/n$ for every $x \in \mathcal{X}$).

Proof. We use Jensen's inequality applied to the strictly concave function log. Let $\mathcal{X}^+ = \{x \in \mathcal{X} : p(x) > 0\}$ denote the support of p, and write $m = |\mathcal{X}^+| \leq n$. By definition,

$$H(X) \;=\; -\sum_{x\in\mathcal{X}^+} p(x)\log p(x) \;=\; \sum_{x\in\mathcal{X}^+} p(x)\log\frac{1}{p(x)}.$$

View the right-hand side as the expectation $\mathbb{E}\Big[\log\frac{1}{p(X)}\Big]$, where the expectation is taken under p. Since log is concave, Jensen's inequality gives

$$\mathbb{E}\Big[\log\tfrac{1}{p(X)}\Big] \;\leq\; \log\Big(\mathbb{E}\Big[\tfrac{1}{p(X)}\Big]\Big).$$

Compute the inner expectation:

$$\mathbb{E}\Big[\tfrac{1}{p(X)}\Big] \;=\; \sum_{x\in\mathcal{X}^+} p(x)\cdot\tfrac{1}{p(x)} \;=\; \sum_{x\in\mathcal{X}^+} 1 \;=\; m.$$

Therefore

$$H(X) \;\leq\; \log m \;\leq\; \log n,$$

where the second inequality uses $m \leq n$ and monotonicity of log.

For the equality condition: log is *strictly* concave, so Jensen's inequality is strict unless $1/p(X)$ is constant on the support, which means $p(x) = 1/m$ for all $x \in \mathcal{X}^+$. Combined with $m \leq n$, equality $H(X) = \log n$ requires $m = n$ and p uniform on $\mathcal{X}$. Conversely, if p is uniform on $\mathcal{X}$, then $H(X) = -\sum_x (1/n)\log(1/n) = \log n$. □

Corollary 5. *For any discrete random variable on n outcomes, $0 \leq H(X) \leq \log n$. The lower bound is achieved by deterministic X, and the upper bound by uniform X.*

5 Joint entropy and subadditivity

Definition 6 (Joint entropy)**.** For discrete random variables X, Y with joint pmf $p(x, y)$,

$$H(X,Y) \;=\; -\sum_{x,y} p(x,y)\log p(x,y).$$

Theorem 7 (Subadditivity)**.** *$H(X,Y) \leq H(X)+H(Y)$, with equality iff X and Y are independent.*

The proof uses non-negativity of KL divergence applied to the joint vs. product of marginals; we defer it to the cross-entropy / KL lesson.

6 Bernoulli entropy as a worked example

Example 8 (Binary entropy function)**.** For $X \sim \mathrm{Bernoulli}(p)$,

$$H(p) \;=\; -p\log_2 p - (1-p)\log_2(1-p).$$

Differentiating, $H'(p) = \log_2\!\left(\frac{1-p}{p}\right)$, which vanishes at $p = 1/2$, giving $H(1/2) = 1$ bit. By Theorem 4 this is the global maximum, as $\log_2 2 = 1$.